

ECE 3441 Fall 2023

Exam 1

Version A

Please do not open the booklet unless instructed to do so, or a substantial penalty will result.

- Closed book, closed notes
- ONE 8.5 x 11" double-sided cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 80 minutes.

First Name Last Name,

Peoplesoft ID,

Date *DD/MM/YYYY*

Question	Points	Maximum
1		15
2		20
3		15
4		20
5		15
6		25

Total: _____/100 possible points + 10 Bonus

Question 1:

Find the **minimum** sum-of-products expression for the following minterm expansion using a 5-variable Karnaugh map. (There is only one correct solution)

$$F_{1A}(A, B, C, D, E) = \sum m(6, 8, 10, 12, 14, 15, 17, 18, 20, 22, 24, 26, 29, 30, 31)$$

BC AE	00	01	11	10
00			1	1
01				
11			1	
10		1	1	1

$A=0$

$$F_{1A} = A'B'E' + A'BCD + A'CD'E' + ADE' + ABCE + AB'CD'E' + AB'CD'E + B'C'D'E'$$

BC AE	00	01	11	10
00		1		1
01	1		1	
11			1	
10	1	1	1	1

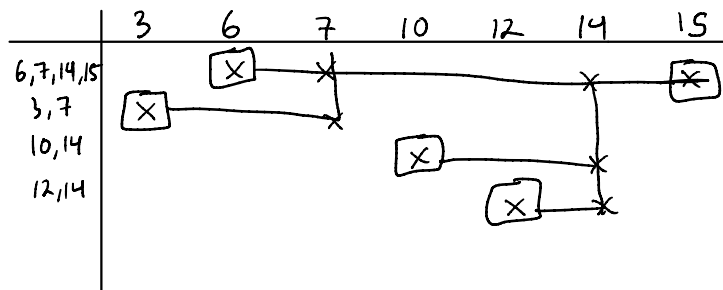
$A=1$

Question 2:

Use the Quine-McCluskey Method followed by the prime implicant chart to find a minimum SOP of the following function. What are the essential prime implicants, if any? (Hint: use a prime-implicant chart)

$$F_{2A}(A, B, C, D) = \sum m(3, 6, 7, 10, 12, 14, 15)$$

3 · 0011	3 0011 ✓	ABCD 3,7 0-11	ABCD 6,7,14,15 -11-
6 · 0110	6 0110 ✓	6,7 011-	6,14,7,15 -11-
7 · 0111	10 1010 ✓	6,14 -110	
10 · 1010	12 1100 ✓	10,14 1-10	
12 · 1100	7 0111 ✓	12,14 11-0	
14 · 1110	14 1110 ✓	7,15 -111 ✓	Prime implicants:
15 · 1111	15 1111 ✓	14,15 111- ✓	A'CD
			ACD'
			ABD'
			BC



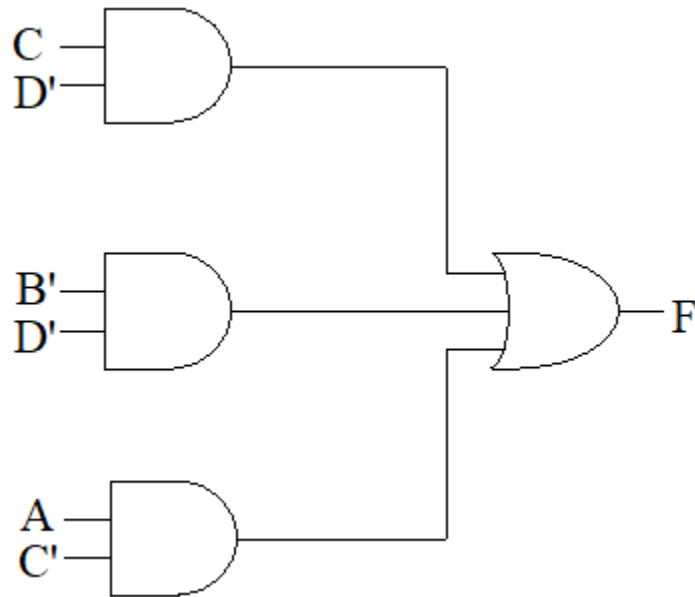
$$F = A'CD + ACD' + ABD' + BC$$

Essential prime implicants

$$A'CD, ACD', ABD', BC$$

Question 3:

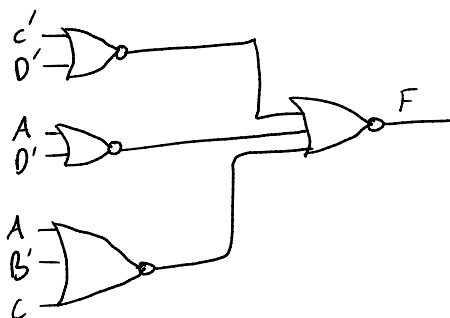
For the following multi-level circuit function F , replace all initial AND/OR gates with NOR gates: (Hint: find F' in SOP form and use DeMorgan's law to find F in POS form)



$$F = CD' + B'D' + AC'$$

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	0	1	1
11	0	0	0	0
10	1	1	1	1

$$F' = (C' + D')(A + D')(A + B' + C)$$



Question 4:

- a) Find the minimum SOP (three terms) of the following function using Boolean algebra:

$$F_{4A1}(A, B, C) = (A'C + BC')' + (A' \oplus BC)$$

State any theorems used in the simplification process.

You must use Boolean algebra to receive any credit.

$$\begin{aligned} F(A, B, C) &= (A+C')(B'+C) + ABC + A'B'C' \quad , \text{Involution Law} \\ &= AB' + \underline{AC} + \underline{B'C'} + \underline{ABC} + \underline{A'B'C'} \quad , \text{10L} \\ &= \overset{y2}{AB'} + \overset{yx}{AC} + \overset{zx}{B'C'} \\ &= AC + B'C' \quad , \text{Absorption Theorem} \\ &\quad , \text{consensus theorem} \end{aligned}$$

$$F_{4A1} = AC + B'C' \quad \times$$

- a) Find the minimum SOP of the following function using Boolean algebra:

$$F_{4A2}(A, B, C) = A + (A + A'B)(A + C) + AB$$

State any theorems used in the simplification process.

You must use Boolean algebra to receive any credit.

$$\begin{aligned} F(A, B, C) &= A + A + AC + A'BC + AB \quad , \text{10L, LOC, IL} \\ &= \underline{A} + \underline{AC} + \underline{A'BC} + \underline{AB} \quad , \text{IL} \\ &= A + A'BC \quad , \text{Absorption Theorem} \end{aligned}$$

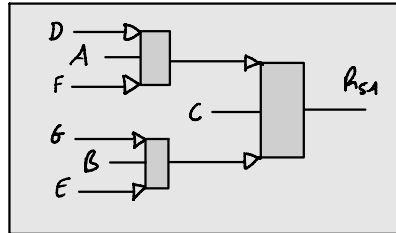
$$F_{4A2} = A + A'BC \quad \times$$

Question 5:

Implement the following function using only tri-state buffers:

$$R_{5A} = ACD + B'C'G + BC'E + A'CF$$

$$R_{5A} = C(AD + A'F) + C'(B'G + BE)$$



Question 6:

Use a 4-to-1 multiplexer and a minimum number of external gates to realize the following function: (Hint: Draw the truth table and you will see how to group the terms together for the 4-to-1 MUX)

$$F_{6A}(A, B, C, D) = \sum m(3, 7, 10, 12, 13, 15) + \sum d(5, 11, 14)$$

Extra Paper